Solar ESS Co. Ltd.
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From sunlight to solar cells

2,700 years of harnessing the sun, and where Solar ESS fits in

Antiquity · passive solar

- **7th c. BC** — Magnifying lenses focus sunlight to start fires.
- **5th–3rd c. BC** — Greeks and Romans orient homes to the south to catch winter sun; Socrates praises the idea.
- **1st–3rd c. AD** — Romans heat bathhouses through large south-facing glass windows.
- **1200s** — Ancestral Puebloans build south-facing cliff dwellings for winter warmth.

Science lays the foundations

- **1767** — Horace-Bénédict de Saussure builds the first solar “hot box” collector, reaching over 100 °C.
- **1839** — Edmond Becquerel, aged 19, discovers the photovoltaic effect: light creates electric current.
- **1860s** — Augustin Mouchot builds solar-powered steam engines.
- **1883** — Charles Fritts makes the first solid solar cell: selenium coated with gold, about 1% efficient.

Birth of modern photovoltaics

- **1905** — Albert Einstein explains the photoelectric effect (Nobel Prize 1921).
- **1941** — Russell Ohl files the patent for the silicon p-n junction solar cell.
- **1954** — Bell Labs (Chapin, Fuller, Pearson) makes the first practical silicon cell, about 6% efficient.
- **1958** — Vanguard 1 becomes the first solar-powered satellite.
- **1970s** — Oil crises push solar research on Earth; costs start to fall.

Solar goes mainstream

- **1980s–2000s** — Grid-connected rooftops and large solar plants scale up with government incentives.
- **2010s** — Panel prices fall by around 90%; solar becomes the cheapest new power in many markets.
- **2025** — Solar makes about 8.8% of the world’s electricity, growing 30% in one year: the fastest-growing source for 21 years in a row.
- **Today** — Back-contact and TOPCon panels pass 24–25% efficiency, about four times Bell Labs’ 1954 cell.

Sources: U.S. Department of Energy solar history timeline; Bell Labs; NASA (Vanguard 1); Ember Global Electricity Review 2026 (solar 2,778 TWh in 2025, +30%).

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Solar in Thailand and our engagement

How solar reached Thai homes, and how Solar ESS is helping it grow

Solar in Thailand

2013 — First rooftop solar feed-in tariff for homes and businesses.

2019 — “People’s Solar” scheme lets households sell surplus power to the grid.

2026 — Tax deduction for home rooftop systems up to 10 kWp connected 3 March 2026 – 31 December 2028; new PEA household scheme buys surplus at 2.20 baht per unit (up to 5 kW).

Our engagement: Solar ESS

2021 — Our partner Thumbs Up ESS is founded (25 March 2021), the start of the network that Solar ESS grows from.

2023 — Solar ESS Co. Ltd. is registered in Chiang Rai (B.E. 2566): “Solar Energy for Sustainable Solution”.

2023–2026 — Projects from north to south: 4 institutional and commercial projects in Trang (620 kWp in total), a 20 kWp commercial rooftop in Chiang Rai, a 10.4 kWp hybrid system with battery, an 8 kWp restaurant in Krabi and an 8 kWp system in Tachileik, Myanmar.

2023–2026 — Offices in Chiang Rai, Krabi, Koh Samui and Nakhon Si Thammarat, and an affiliated network of solar businesses across Thailand.

2026 — New website www.solaress.org goes live with a free technical library, comparison reports and an installer Jobs platform.

Next — Helping more homes, farms and businesses in Thailand run on their own clean energy, with safe design, Tier-1 equipment and honest advice.

☀️ **Solar Energy for Sustainable Solution.** From Becquerel’s 1% discovery to today’s 25% panels, each step made solar cheaper, safer and easier to own. Our job is to bring that progress to your roof, done properly.

Thai policy dates: Energy Regulatory Commission and PEA announcements. Project sizes from Solar ESS records; customers are not named.